

CO1-AT3

1. Student Management System:

```

class Student {
    private String name;
    private int regNo;
    private int marks;

    void studentDetails(String n, int r, int m) {
        name = n;
        regNo = r;
        marks = m;
    }

    int calcAverage() {
        return marks;
    }

    void displayResults() {
        if (calcAverage() >= 50)
            System.out.println("PASS");
        else
            System.out.println("FAIL");
    }
}

```

85/100

Library Book Management:

```

class Book {
    private String title;
    private String author;
    private int ISBN;
    private boolean isAvailable = true;
}

```



```
void issueBook () {
```

```
    if (C.isAvailable)
```

```
    {
```

```
        isAvailable = false;
```

```
        System.out.println("Book Issued");
```

```
    }  
    void returnBook () {
```

```
        isAvailable = true;
```

```
        System.out.println("Book returned");
```

```
    }  
    String getTitle () {
```

```
        return title;
```

```
    }  
    boolean getAvailability () {
```

```
        return isAvailable;
```

```
    }  
}
```

3. Employee salary calculation:

```
class Employee {
```

```
    private int empID;
```

```
    private String name;
```

```
    private double basicPay;
```

```
    double calcGrossSalary () {
```

```
        double hra = basicPay * 0.20;
```

```
        double da = basicPay * 0.10;
```

```
        return basicPay + hra + da;
```

```
    }  
}
```



```
double calcNetSalary() {
```

```
    double gross = calcGrossSalary();
```

```
    double pf = gross * 0.05;
```

```
    double tax = gross * 0.10;
```

```
    return gross - pf - tax;
```

```
}
```

```
void displayPaySlip() {
```

```
    System.out.println("Gross Salary : " + calcGrossSalary());
```

```
    System.out.println("Net Salary : " + calcNetSalary());
```

```
}
```

```
}
```

A. Shape Area Calculator:

```
class shape {
```

```
    void calculateArea() {
```

```
        System.out.println("Area");
```

```
}
```

```
class circle extends shape {
```

```
    void calculateArea() {
```

```
        double radius = 5;
```

```
        System.out.println(3.14 * radius * radius);
```

```
}
```

```
class rectangle extends shape {
```

```
    void calculateArea() {
```

```
        int length = 10;
```

```
        int breadth = 5;
```

```
        System.out.println(length * breadth);
```

```
}
```

```
}
```



```
class Triangle extends Shape {
```

```
void calculateArea() {
```

```
int base = 10;
```

```
int height = 6;
```

```
System.out.println ((base * height) / 2);
```

```
}
```

```
}
```

```
5. Vehicle Registration System:
```

```
class Vehicle {
```

```
String regNo;
```

```
String owner;
```

```
String fuelType;
```

```
void displayDetails() {
```

```
System.out.println (regNo);
```

```
System.out.println (owner);
```

```
System.out.println (fuelType);
```

```
}
```

```
}
```

```
class TwoWheeler extends Vehicle {
```

```
int engineCapacity;
```

```
void displayDetails() {
```

```
super.displayDetails();
```

```
System.out.println (engineCapacity);
```

```
}
```

```
}
```

```
class FourWheeler extends Vehicle {
```

```
int seatingCapacity;
```

```
boolean acStatus;
```



```
void displayDetails() {
```

```
    super.DisplayDetails();
```

```
    System.out.println("Seating Capacity");
```

```
    System.out.println("acstatus");
```

```
}
```

```
}
```

6. Bank Account Hierarchy;

```
class BankAccount {
```

```
    String accNo;
```

```
    String holder;
```

```
    double balance;
```

```
    void withdraw(double amount) {
```

```
        if (balance >= amount)
```

```
        {
```

```
            balance = balance - amount;
```

```
            System.out.println("Amount withdrawn");
```

```
        } else
```

```
        {
```

```
            System.out.println("Insufficient Balance");
```

```
        }
```

```
    }
```

```
class SavingsAccount extends BankAccount {
```

```
    double calcInterest()
```

```
    {
```

```
        return balance * 0.05;
```

```
    }
```

```
    void withdraw(double amount) {
```

```
        if (balance >= amount)
```

```
        {
```

```
            balance = balance - amount;
```

```
            System.out.println("Savings withdraw success");
```

```
        }
```



```

else
{
    System.out.println("cannot withdraw");
}
}
}
}

class currentAccount extends BankAccount {
    double overdraft = 5000;
    void withdraw (double amount) {
        if (balance + overdraft >= amount)
        {
            balance = balance - amount;
            System.out.println("withdrewd using overdra");
        }
        else
        {
            System.out.println("overdraft limits exceeded");
        }
    }
}

```

7. Hospital Record System:

```

class Patient {
    private String diagnosis;
    private String prescription;
    void generateBill() {
        System.out.println("Patient Bill");
    }
}

class InPatient extends Patient {
    int wardNo;
    int days;
    double dailyCharge = 1000;
}

```



```
void generateBill () {
```

```
    System.out.println (days * dailyCharge);
```

```
}  
class Outpatient extends Patient {
```

```
    String appointmentDay;
```

```
    double consultationFee = 500;
```

```
    void generateBill () {
```

```
        System.out.println (consultationFee);
```

```
}  
8. class Product {
```

```
    String productName;
```

```
    double price;
```

```
    void applyDiscount () {
```

```
        System.out.println (price);
```

```
}  
class Electronics extends Product {
```

```
    void applyDiscount () {
```

```
        System.out.println (price - (price * 0.10));
```

```
}  
class Clothing extends Product {
```

```
    void applyDiscount () {
```

```
        System.out.println (price - (price * 0.20));
```

```
}  
class Groceries extends Product {
```

```
    void applyDiscount () {
```

```
        System.out.println (price - (price * 0.05));
```


9. class Staff {

protected String name;

protected int staffID;

void calcAllowance () {

System.out.println ("Base Allowance");

class TeachingStaff extends Staff {

void calcAllowance () {

System.out.println ("Teaching Allowance");

class Professor extends TeachingStaff {

void calcAllowance () {

System.out.println ("Professor Allowance");

class Lecturer extends TeachingStaff {

void calcAllowance () {

System.out.println ("Lecturer Allowance");

class NonTeachingStaff extends Staff {

void calcAllowance () {

System.out.println ("Non-Teaching Allowance");

10. Smart Home Device:

```
class SmartDevice {
```

```
    void executeCommand (String cmd) {
```

```
        System.out.println (cmd);
```

```
    }
```

```
}
```

```
class SmartLight extends SmartDevice {
```

```
    void executeCommand (String cmd) {
```

```
        System.out.println ("Light: " + cmd);
```

```
    }
```

```
}
```

```
class SmartLock extends SmartDevice {
```

```
    private int pin = 1234;
```

```
    void executeCommand (String cmd) {
```

```
        System.out.println ("Lock: " + cmd);
```

```
    }
```

```
}
```

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